

ops-profiler

A warehouse-design analytics engine. Raw WMS data in. A sourced design package — system sizing and a headcount ROI model — out.

The Problem

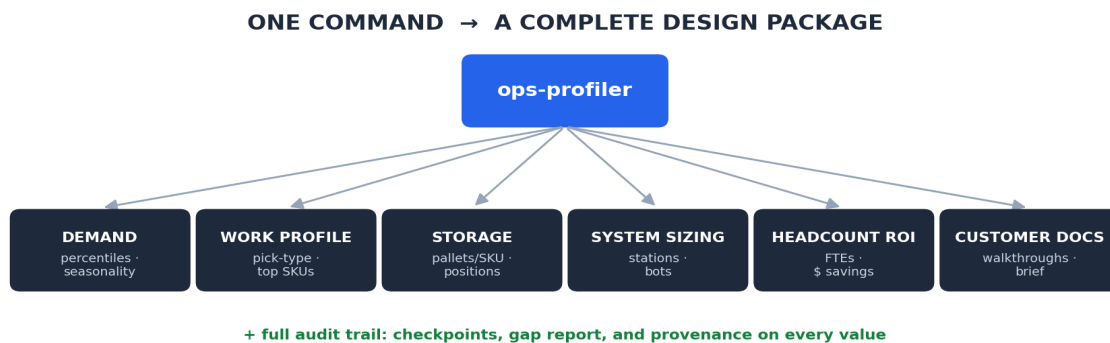
To evaluate a warehouse for automation, a solutions engineer must answer hard quantitative questions from whatever data the customer provides: peak order volume, what the picking work looks like, how many stations and bots the system needs, and the question that closes the deal — how much labor cost it displaces. By hand this was about a week of senior-analyst work per engagement, it varied by who did it, and the provenance vanished when the workbook closed.

The Approach

A pipeline that goes all the way from raw files to a sourced design package, at any data richness level. Every value carries a source pointer, so the whole package is auditable.



What one command produces



The Differentiators

It reaches a financial decision, not just a profile. It keeps going through first-principles system sizing into a headcount ROI model — it produced the savings figure that justified the purchase, every input sourced.

First-principles sizing with the math exposed. Station and bot counts derive from line volume, cycle times, and stated assumptions — never a template. A reviewer can see exactly how a recommendation was reached and challenge any input.

Deterministic core, AI only at the checked edges. Parsing, percentile math, and sizing are deterministic. The AI maps messy headers and reads prose, and its output is always checked against the schema. The AI proposes; the rules dispose. It never touches the math — that is what makes the output defensible.



A hallucinated field is rejected before it can enter the design package.

The Payoff

~a week	one command	every value	stations · bots · ROI
of analyst work, by hand	automated	permanently traceable	sized and sourced

Used on live customer engagements, including a deal where the pricing landed exactly where the customer expected.

Python · DuckDB · pandas/NumPy · deterministic core · Claude (checked AI edges) · provenance tracking
John White · AI Operations & Optimization · [linkedin.com/in/johnwhiteimse](https://www.linkedin.com/in/johnwhiteimse)